Spherity Strategy and Market Positioning Update

Digital Product Passports, European Business Wallets and the agentic internet

Dr. Carsten Stöcker
Spherity GmbH

Prepared: August 2026

Abstract

Spherity combines product identity and lifecycle data with verified enterprise identity, authority and access control. Spherity operationalises this architecture through VERA for Digital Product Passports, EIDA for European Business Wallets and CARO for regulated enterprise verification. Together, these capabilities create a cyber-secure foundation for regulated supply chains, circular business models and machine-to-machine and agentic transactions.

Keywords. Digital Product Passport, European Business Wallet, verifiable credentials, agentic commerce, product compliance

1 Market position

Gartner® published the Emerging Market Quadrant for Digital Product Passport — Established Vendors on 6 July 2026. Spherity is positioned as the sole vendor in the Pioneers quadrant. [1]

Spherity sees this identity-first position as the foundation for a wider trust architecture that can extend from physical-product DPPs to AI Service Passports for AI systems and services. This extension is Spherity's strategic outlook and was not part of Gartner's DPP evaluation.

Spherity treats the Gartner placement as one independent market perspective. Spherity's own view is that its technology leadership in identity-first and cyber-secure DPP infrastructure rests on its architecture, regulated production experience and partner ecosystem. The strategic objective is to make product data trustworthy, usable across company boundaries and suitable for automated decisions.

2 DPP and EBW convergence: one architecture, two planes

A DPP provides product attributes at product, SKU, batch and serial-number levels. In addition, it must uniquely identify the product, connect data from multiple organisations and preserve its meaning over many years. It must also answer a second set of questions: Who issued a statement? Is the issuer a valid legal entity? Was the person, system or AI agent authorised to act? Which recipients may access or update the data? Is the evidence still current?

Spherity therefore combines two complementary planes:

- **The data plane.** The DPP carries or resolves structured product data, material and component information, certificates, provenance, manufacturing and service events, software status, sustainability indicators and end-of-life evidence. Verifiable credentials support data integrity, semantic meaning, and machine verification.
- **The control plane.** The European Business Wallet (EBW) establishes trusted enterprise identity, market roles, licences, mandates and powers of representation. It supports policy-based access, selective disclosure, credential status and auditable delegation to employees, machines and AI agents.

The same control plane can secure access to the EU DPP Registry, which became operational in July 2026, and to sector-specific registries. In Spherity's architecture, the EBW can verify the economic operator or authorised representative, prove the mandate under which a person, service provider or AI agent acts, and support policy-controlled registration or update of product and operator identifiers. The Registry remains the common EU index. The EBW provides verifiable legal identity and authority at its organisational interface. [2]

Together, both planes form a verifiable and access-controlled DPP. A recycler can prove eligibility before receiving restricted disassembly data. A regulator can verify who submitted a compliance statement. A service provider can update a maintenance event within a defined mandate. An AI agent can act only within a narrow, revocable authority. The architecture protects data sovereignty while allowing the evidence needed for compliance and circular operations. In Spherity's portfolio, VERA operationalises the DPP data plane and EIDA provides reusable enterprise-identity, credential and access-control capabilities for the control plane; both remain modular and interoperable with existing systems and data ecosystems.

3 Execution: batteries first, then horizontal scale

Spherity is executing customer programmes across traction batteries, commercial vehicles and public transport fleets, battery energy storage systems (BESS), and light means of transport (LMT) batteries. LMT is the legal term used in the EU Battery Regulation for batteries in vehicles such as e-bikes and e-scooters. [3] These programmes cover product onboarding, supplier data collection, regulatory data models, lifecycle events, analytics and integration with enterprise systems.

The battery market is the first large-scale proving ground. It combines strict regulatory deadlines, high-value assets, safety obligations, complex bills of materials and long service lives. It also creates measurable value beyond compliance. Fleet operators can improve maintenance. Manufacturers can support warranty and recall processes. Second-life operators and recyclers can make better decisions using verified state, provenance and handling information.

Spherity's execution capability also builds on CARO, its production master-data and authorised-trading-partner service for the regulated US pharmaceutical supply chain. CARO demonstrates the operational disciplines required for trust infrastructure: verified enterprise identity, current licence and eligibility checks, interoperable credentials, API integration, status monitoring and auditability. This production experience provides a reusable delivery pattern for moving VERA and EIDA from pilots into managed compliance services.

The next step is a horizontal expansion of the same trust components. Current work includes industrial products and steel with Manufacturing-X project partners, energy assets with energy data-X project partners, and chemicals with Chem-X project partners. Spherity is also extending the model to consumer products. The sector data models differ. The trust functions remain stable: product identity, legal-entity identity, provenance, authority, access policy and verifiable lifecycle evidence. The scaling model should separate this reusable horizontal trust core from sector-specific data models and workflows, then expand through implementation, co-selling and reseller partnerships.

4 Automating compliance, product safety and cybersecurity across adjacent regulatory regimes

A DPP can become a reusable evidence and workflow layer across related EU product, chemical, safety, AI and cybersecurity rules. It does not replace the accountable economic operator, conformity assessment, technical documentation or regulatory reporting. It can reduce repeated data collection by linking each product to current, verifiable evidence and by automating controlled updates, status checks and notifications.

- **General Product Safety Regulation.** Links product, batch or serial identity with the responsible economic operator, safety warnings, incident records, recall notices and consumer remedies. [4]
- **Product Liability Directive.** Preserves evidence about the product version, software updates, warnings, modifications and lifecycle state that can help establish what was placed on the market and how it changed over time. [5]
- **REACH and CLP.** Carries substance, substance of very high concern, hazard, safe-use and supplier evidence with controlled access and traceable updates across the supply chain. [6]
- **Detergents and Surfactants Regulation.** Supports the mandated DPP, digital labelling, product and operator identifiers, Registry interaction and customs verification. [7]
- **Machinery Regulation.** Connects conformity and technical documentation, safety instructions, maintenance, modifications, software versions and AI-related safety evidence for machinery and related products. [8]
- **EU AI Act.** An AI Service Passport can structure provider and operator identity, intended purpose, model and version, TEVV, logs, post-market monitoring and incident evidence for AI systems. [9]
- **Cyber Resilience Act.** Supports product versioning, vulnerability status, secure updates, support periods, incident evidence and auditable cybersecurity maintenance for products with digital elements. [10]
- **Construction Products Regulation.** Aligns declarations of performance and conformity, safety information, technical documentation and access rights through the construction DPP system. [11]

The common architecture is a product or service identity linked to authoritative organisations, verifiable evidence, controlled update rights and event-driven business logic. This reduces duplicate data collection and allows the same compliance event to serve regulators, customers, service networks and internal risk systems.

5 Cyber-secure DPPs for the agentic internet

CIRPASS-2 is a European initiative supporting the development of interoperable Digital Product Passports. Its report, Risks and Mitigations: A Companion to D4.1 Reference Architecture, identifies key data, security and governance risks and treats their mitigation as a core architectural requirement rather than a later add-on. [12]

This is critical because DPPs will increasingly be read, interpreted and acted upon by software systems and AI agents. A weak passport can spread false claims at machine speed. An insecure update path can distort maintenance, repair or recycling decisions. An unidentified issuer can create false provenance. Excessive disclosure can expose trade secrets and sensitive supply-chain relationships.

Spherity's cyber-security approach combines cryptographic product and enterprise identity, verifiable credentials, selective disclosure, credential status and revocation, policy-based access, tamper-evident event histories and signed audit evidence. The EBW adds proof of legal identity and authority. For AI-mediated decisions, Spherity's evidence-graph direction also preserves source, issuer, transformation, validation, freshness, conflicts and uncertainty. Together, the control plane establishes whether an actor is authorised, while the data plane establishes whether the evidence supporting the action is fit for purpose. This allows a system to verify not only that data is intact, but also that the issuer was entitled to make the statement and that the relying party is entitled to receive it.

These controls become essential in the agentic internet. Procurement agents, service agents, customs systems, regulators and circular-economy platforms will negotiate, inspect and act on product data. Machine-readable information alone is insufficient. Agents need machine-verifiable trust, purpose limits and accountable delegation. A product passport can then serve as a trusted interface between physical products, organisations and automated processes.

6 Strategic outlook: AI Service Passports for Agentic Commerce, Industrial AI and Physical AI

Spherity's strategic outlook extends the DPP and EBW architecture to AI systems and services through an AI Service Passport (AISP). This is a forward-looking research and product direction rather than a current regulatory requirement. Spherity proposes the AISP as a DPP-like, verifiable lifecycle record for an AI service. It links the service to its provider or operator and records provenance, model family and version, configuration and guardrails, testing, evaluation, verification and validation (TEVV), operational controls, incidents and updates.

The EBW provides verified legal-person identity, market roles and revocable powers of attorney for people, systems and agents acting on behalf of an organisation. The AISP provides the AI service's provenance and safety profile. Together, they allow a relying party or another agent to verify who is responsible, what the AI is authorised to do, which service version is running and which evidence supports its use. This can support policy-based allow, restrict or block decisions and continuous third-party risk management.

- **Agentic Commerce.** Verifies the organisation behind an agent, its mandate, transaction scope and service provenance before an agent negotiates, purchases, pays or commits an organisation.
- **Industrial AI.** Supports cross-company AI services in supply chains and data spaces through authenticated roles, usage policies, provenance, TEVV evidence and auditable

agent-to-agent interaction.

- **Physical AI.** Links AI models and versions, safety controls, human oversight, operational limits, updates and incidents to the machinery, vehicle, robot or energy asset on which the AI acts.

An AISP supports AI Act documentation, conformity assessment, functional-safety engineering or organisational controls. It also makes evidence portable, current and machine-verifiable across organisational boundaries. The DPP governs the physical product and its lifecycle evidence. The AISP governs the AI service that observes, decides or acts. The EBW anchors legal responsibility and authority across both.

7 The micro-ERP: a product-centric execution layer

Spherity concept. Each product can become a governed micro-ERP: a DPP combined with bounded lifecycle business logic, identity and security. The term describes a product-centric execution layer, not a replacement for enterprise systems. The product becomes a persistent unit of data, rights and process across organisational boundaries.

Traditional ERP, MES, PLM and PIM systems organise data around one company, one plant, one engineering environment or one catalogue. The product lifecycle crosses all of them. It also crosses suppliers, owners, service providers, authorities, resale platforms and recyclers. One-step-up, one-step-down transparency may satisfy a limited traceability need, yet it does not create a continuous product record or a reusable control model.

The micro-ERP concept changes the unit of coordination. The product identity resolves to its current state, evidence, obligations, permissions and authorised services. Business logic can determine which data must be requested, which party may update a record, when a certificate expires, when a safety event triggers a notification, or which end-of-life path is permitted. Security controls remain attached to the product context throughout ownership and use changes.

This approach does not require companies to replace ERP, MES, PLM or PIM systems. Those systems remain important systems of record and execution. The DPP becomes a distributed product-centric layer above and between them. It creates pressure on incumbent providers because lifecycle data, access control and inter-company process logic can no longer remain confined to one enterprise application. Providers will need open interfaces, verifiable data and cross-ecosystem identity support.

8 Strategic priorities for 2026 to 2029

Spherity's strategy follows five linked priorities. First, industrialise VERA deployments and supplier onboarding for regulated battery production. Second, integrate VERA and EIDA into a repeatable two-plane offering for DPP Registry and cross-company workflows. Third, reuse identity, credential, policy, integration and evidence-graph components across product sectors and European data spaces. Fourth, scale delivery through implementation, co-selling and reseller partners. Fifth, develop agent-ready lifecycle services and crypto-agile, post-quantum-resilient identity and trust services once trustworthy product and enterprise identities are in place.

This sequence is deliberate. Compliance creates the initial demand and a defined minimum data set. Operational services create recurring value. The trust layer makes both scalable across firms and jurisdictions. In this model, a DPP is not a document attached to a product.

It is a secure product interface for compliance, lifecycle operations and automated commerce. Execution should be measured by conversion from pilots to production, supplier-onboarding time, recurring compliance-service revenue, partner-sourced deployments, cross-sector reuse of core components and the proportion of transactions that verify both evidence and authority.

We believe that the Gartner placement provides useful external context for this direction. Spherity is positioned as the sole Pioneer in the 2026 established-vendor DPP EMQ. Spherity views this placement as consistent with its focus on identity-first infrastructure and verifiable data exchange. Market leadership will still depend on execution: converting customer programmes into repeatable deployments, expanding delivery capacity and forming strong implementation partnerships across sectors and regions.

9 Source attribution and legal notice

References

[1] Gartner, *Emerging Market Quadrant for Digital Product Passport — Established Vendors*, Kevin Lawrence, Marta Muñoz and Rajan Saini, 6 July 2026, ID G00850583. Gartner document 8098797.

[2] European Commission, *The Digital Product Passport Registry is now live*, 20 July 2026. Commission news announcement; Regulation (EU) 2024/1781 establishing a framework for ecodesign requirements for sustainable products. EUR-Lex.

[3] Regulation (EU) 2023/1542 concerning batteries and waste batteries, Article 3. EUR-Lex.

[4] Regulation (EU) 2023/988 on general product safety. EUR-Lex.

[5] Directive (EU) 2024/2853 on liability for defective products. EUR-Lex.

[6] Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), and Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures (CLP). EUR-Lex: REACH; CLP.

[7] Regulation (EU) 2026/405 on detergents and surfactants. EUR-Lex.

[8] Regulation (EU) 2023/1230 on machinery. EUR-Lex.

[9] Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence. EUR-Lex.

[10] Regulation (EU) 2024/2847 on horizontal cybersecurity requirements for products with digital elements (Cyber Resilience Act). EUR-Lex.

[11] Regulation (EU) 2024/3110 laying down harmonised rules for the marketing of construction products. EUR-Lex.

[12] CIRPASS-2 Consortium, *Risks and mitigations: a companion to D4.1 Reference Architecture*, version 1.1, 5 June 2026, DOI 10.5281/zenodo.20670585. Zenodo record.

Third-party rights

Gartner Objectivity Disclaimer